

HR Analytics & Employee Attrition — Case Study

1. Business Problem

Employee attrition is a major cost driver for any company — recruiting, onboarding, and lost productivity add up fast when turnover is high. This project analyzes **1,470** employee records to answer the question every HR/People Analytics team is asked: **who is leaving, and why** — with enough specificity to act on, not just a single overall percentage.

2. Approach

- Analyzed employee attrition using Python to identify patterns across **department, job role, age group, tenure, and overtime status**.
- Used a **chi-square test** to validate whether overtime is significantly associated with employee attrition instead of relying only on percentage differences.
- Also analyzed the relationship between attrition and numerical factors such as **job satisfaction, work-life balance, monthly income, and years at the company**.
- Built a **2-page Power BI dashboard** with an overview page and a driver analysis page, including the **Key Influencers** visual to independently identify the main factors associated with attrition.

3. Key Findings

Metric	Value	So what
Overall attrition rate	16.12% (1,470 employees)	About 1 in 6 employees leave, making retention a key business concern.
Attrition — Over Time = Yes	30.53%	Employees working overtime have nearly 3× higher attrition.
Attrition — Over Time = No	10.44%	Employees without overtime show substantially lower attrition.
Highest-risk job role	Sales Representative, 39.76%	Sales Representatives have the highest attrition and need targeted retention efforts.
Highest-risk age band	18–25, 35.77%	Early-career employees leave at a much higher rate.
Highest-risk tenure band	0–2 years, 29.82%	The first two years are the most critical period for employee retention.
Top Segment (Key Influencers)	59.6% attrition, 99 employees	A specific employee segment has significantly higher risk and can be targeted for retention.

4. Recommendations

- **Review overtime workload:** Examine workload and staffing for employees working overtime, particularly in Sales, as overtime is the strongest identified factor associated with attrition.
- **Strengthen early-career support:** Improve onboarding, mentorship, and employee engagement during the first two years, when attrition is highest.
- **Target high-risk employees:** Focus retention efforts on the identified high-risk segment instead of applying broad company-wide initiatives.
- **Investigate Sales Representative attrition:** Review factors such as workload, compensation, sales targets, and management practices to understand why this role has significantly higher attrition.